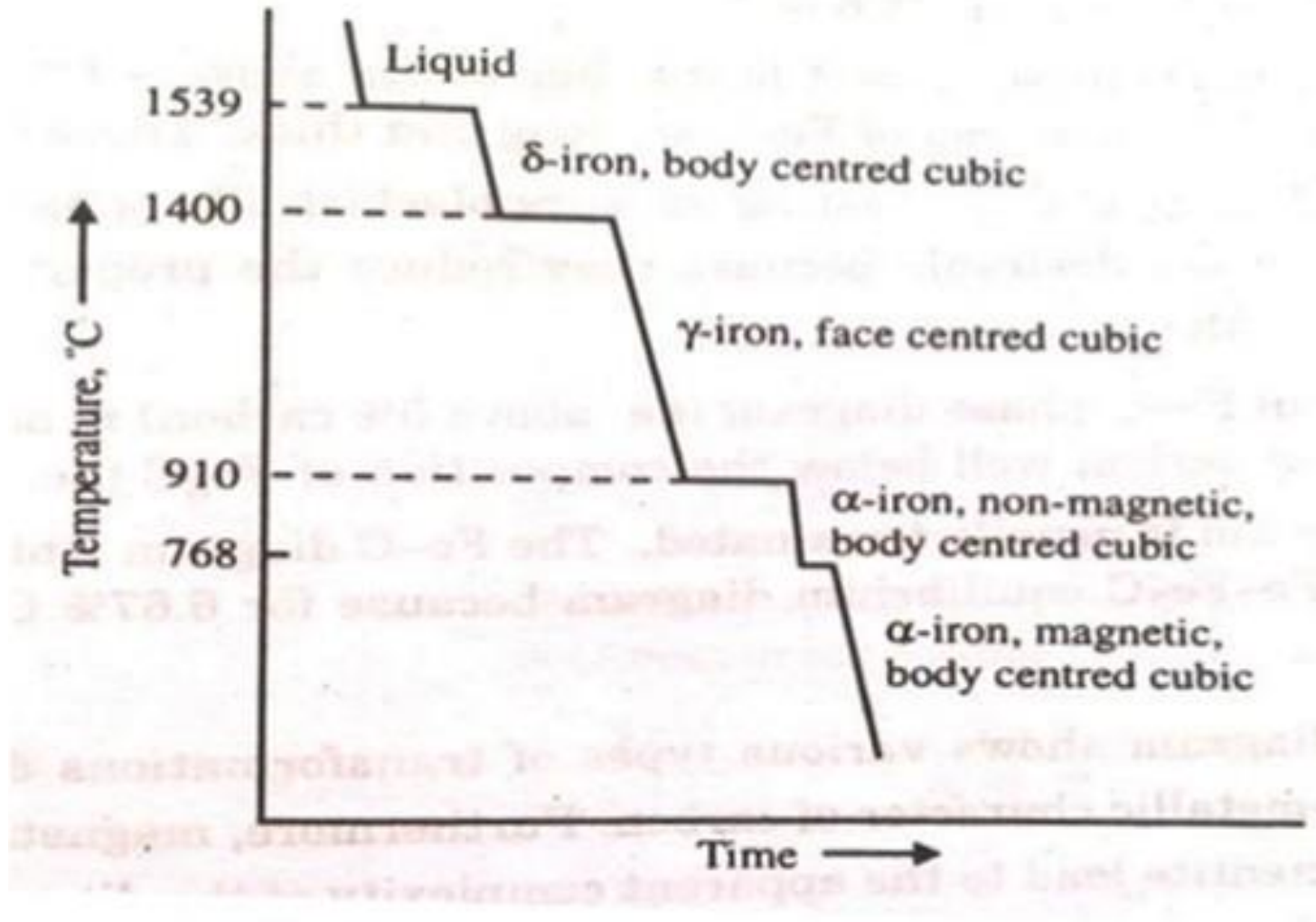


Fe_C Diagram

- Pure iron exist in more than one crystal structure in solid state.
- Soft iron is the best known example for allotropy of iron, which can be strengthened/hardened by adding small amount of other metals.
- Addition of carbon to soft iron can increase the hardness of soft iron by 1000 times, the alloy is called as steel.
- There are three allotropic forms of iron α (BCC), γ (FCC), and δ (BCC)
- When liquid iron is cooled down, it crystallizes at 1538 °C to
 - δ (BCC) iron.
- Upon further cooling, the crystal structure gets converted to γ (FCC) at 1394 °C.
- The crystal structure changes into α -phase (BCC) at 912°C .
- These allotropes are treated as different phases.

Allotropic forms



- Family of iron carbon alloys having carbon content upto 2.14% is called as steel.
- Steels are solid solutions of iron and carbon
- If the carbon content is more than 2.14% then it is called as cast iron.
- In cast irons most of the carbon exists as inter metallic compound, iron carbide.
- In phase diagram, the allotropic forms of iron forms different solid solutions which also can be considered as phases.
- The intermetallic compound, iron carbide, is also a phase present in the system.

α ferrite

- BCC α ferrite is also an interstitial solid solution of carbon in BCC iron.
- In the BCC α ferrite, only small concentrations of carbon are soluble; the maximum solubility is 0.022 wt% at 727°C and at room temperature it is reduced to 0.008%.
- The limited solubility is explained by the shape and size of the BCC interstitial positions, which make it difficult to accommodate the carbon atoms.
- Even though present in relatively low concentrations, carbon significantly influences the mechanical properties of ferrite.
- This particular iron-carbon phase is relatively soft, may be made magnetic at temperatures below 768°C.

Austenite FCC

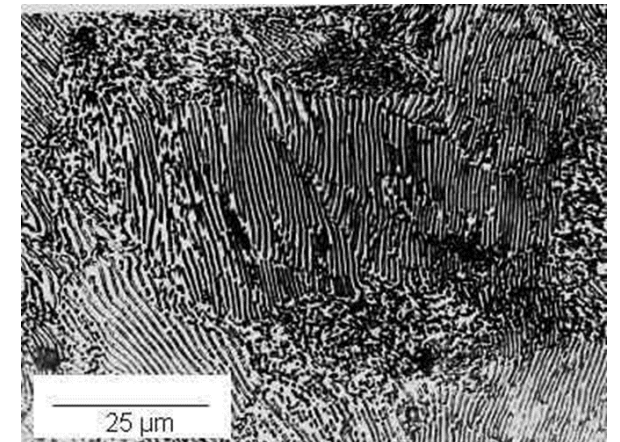
- The austenite, or γ phase of iron (FCC), when alloyed with carbon alone, is not stable below 727°C.
- The maximum solubility of carbon in austenite, 2.14 wt%, occurs at 1147°C. This solubility is approximately 100 times greater than the maximum for BCC ferrite.
- Stable between 912-1394°C
- The FCC interstitial positions are larger, and, therefore, the strains imposed on the surrounding iron atoms are much lower.
- The phase transformations involving austenite are very important in the heat treating of steels.
- Austenite is non-magnetic.
- The α and δ ferrite is virtually the same as ferrite, except for the range of temperatures over which each exists.

Cementite or Fe_3C or iron carbide_ORTHORHOMBBIC

- Cementite (Fe_3C) forms when the solubility limit of carbon in ferrite is exceeded 0.002% below 727°C (for compositions within the $\alpha + \text{Fe}_3\text{C}$ phase region).
- Fe_3C will also coexist with the phase between 727 and 1147°C).
- Mechanically, cementite is very hard and brittle; the strength of some steels is greatly enhanced by its presence.
- Strictly speaking, cementite is only metastable; that is, it will remain as a compound indefinitely at room temperature.
- However, if heated to between 650 and 700°C for several years, it will gradually change or transform into a iron and carbon, in the form of graphite, which will remain upon subsequent cooling to room temperature.
- It may be noted that one eutectic exists for the iron-iron carbide system, at 4.30 wt% C and 1147°C ;

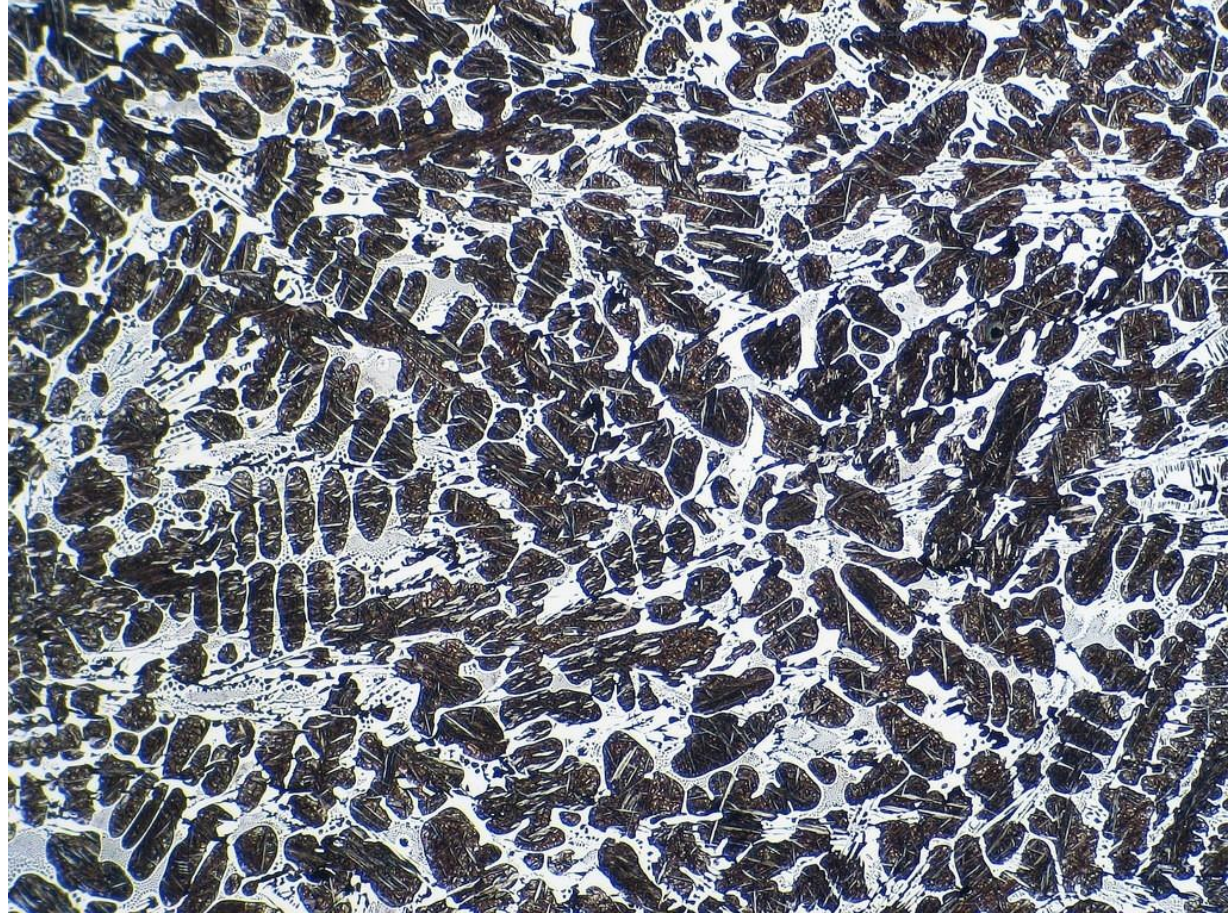
Pearlite structure

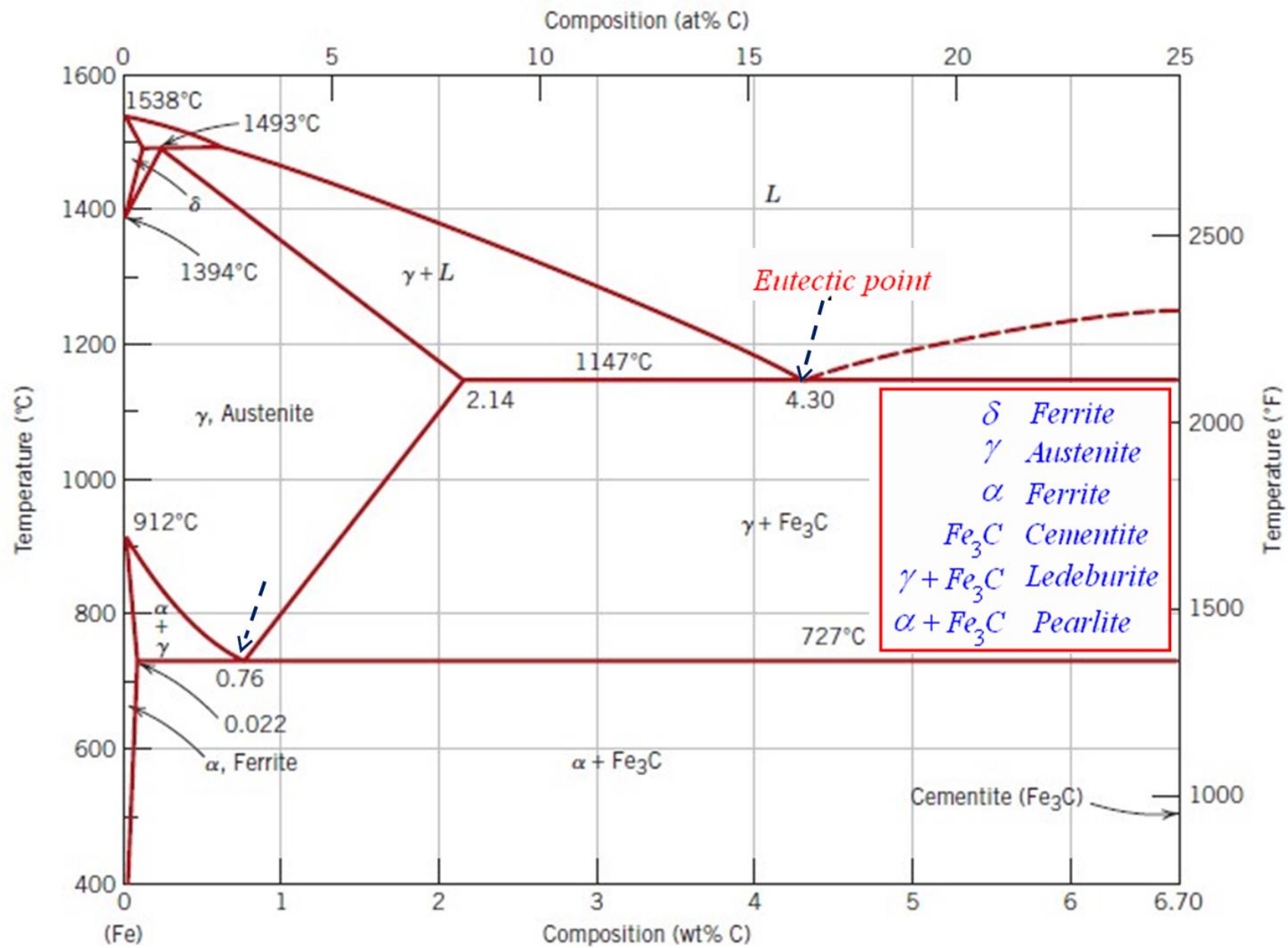
- A laminated structure formed of alternate layers of ferrite and cementite with average composition 0.83% carbon
- Pearly lustre in the microscope
 - Interference of light in its regular layers
- Most common constituent of steel
- It combines the hardness and strength of cementite with the ductility of ferrite and is the key to the wide range of the properties of steel
- The laminar structure also act as barrier to crack movement as in composites. This gives it toughness.



ledeburite

- **ledeburite** is a mixture of 4.3% carbon in iron and is a eutectic mixture of austenite and cementite.



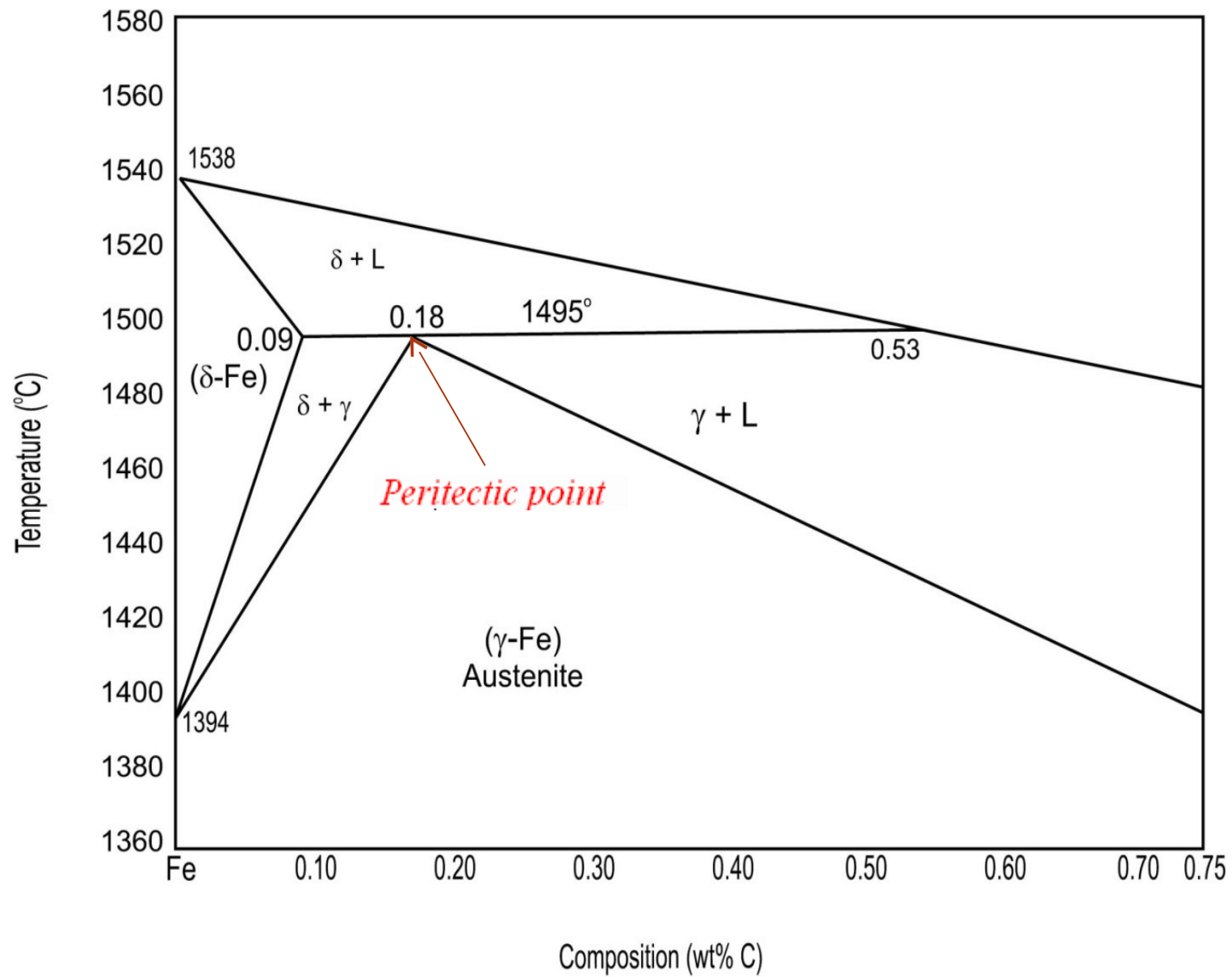




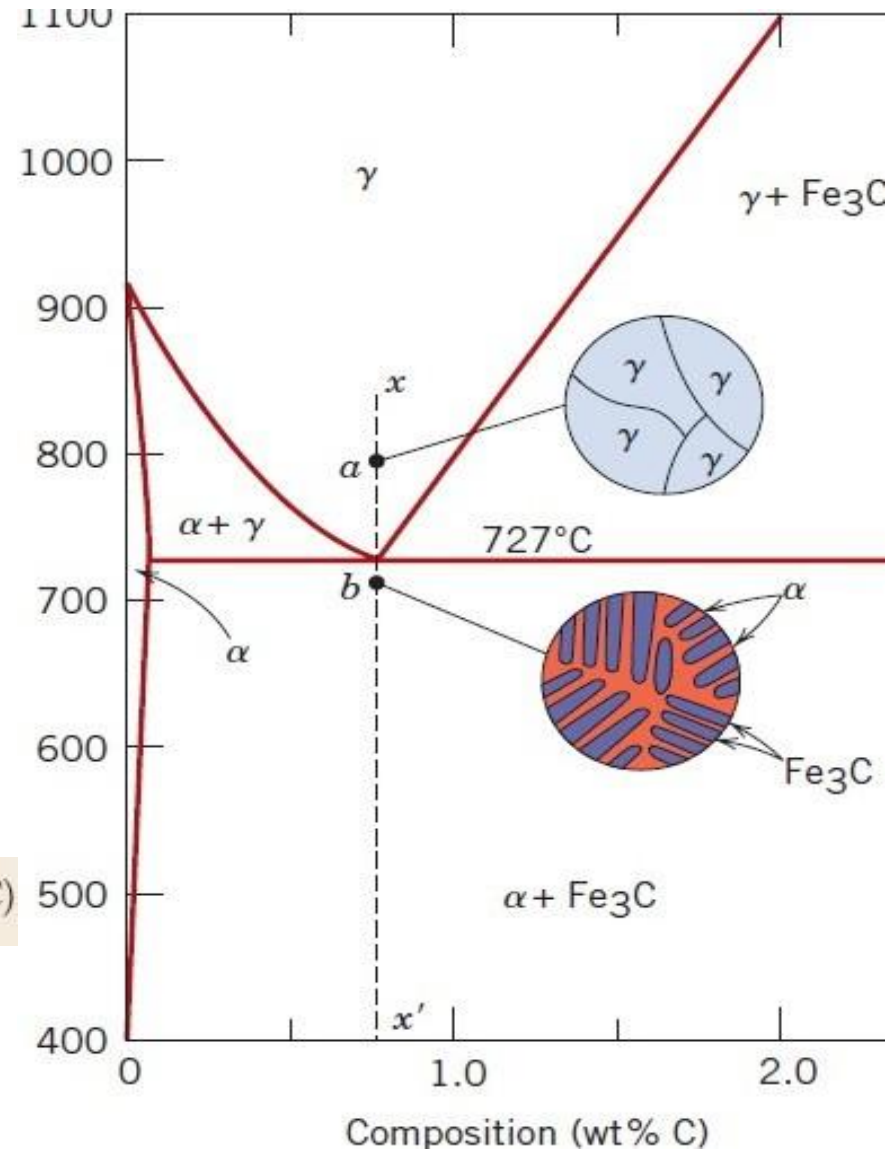
- Peritectic reaction at 1495°C and 0.18%C,
 $\delta\text{-ferrite} + L \leftrightarrow \gamma\text{-iron (austenite)}$

Eutectic reaction at 1147°C and 4.3 %C,
➤ $L \leftrightarrow \gamma\text{-iron} + \text{Fe}_3\text{C (cementite)}$
[ledeburite]

Eutectoid reaction at 727°C and 0.77%C,
 $\gamma\text{-iron} \leftrightarrow \alpha\text{-ferrite} + \text{Fe}_3\text{C (cementite)}$
[pearlite]



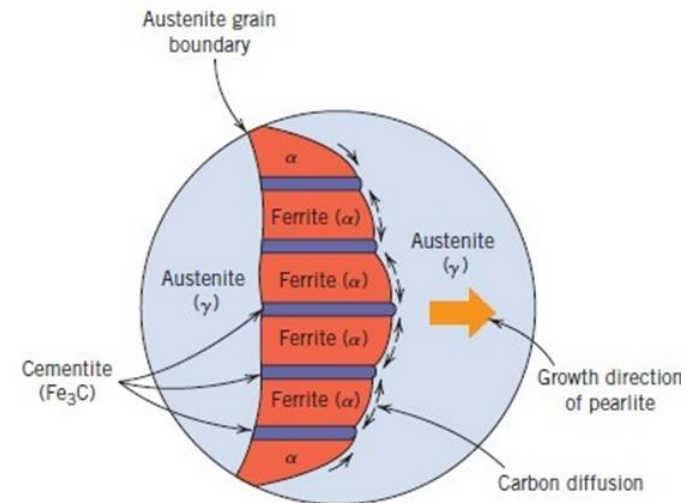
- Consider, an alloy of eutectoid composition (0.76 wt% C) is cooled from a temperature within the γ phase region, say, 800°C—that is, beginning at point *a* and moving down the vertical line *xx'*.
- Initially, the alloy is composed entirely of the austenite phase having a composition of 0.76 wt% C.
- As the alloy is cooled, there will be no changes until the eutectoid temperature (727°C) is reached.
- Upon crossing this temperature to point *b*, the austenite transforms



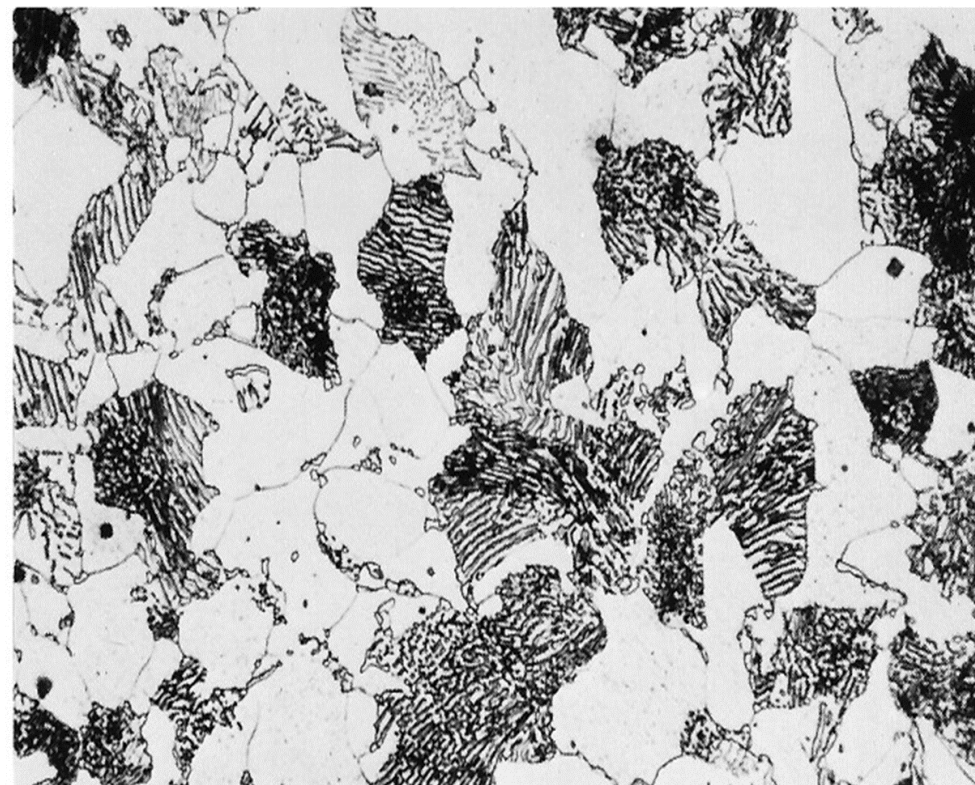
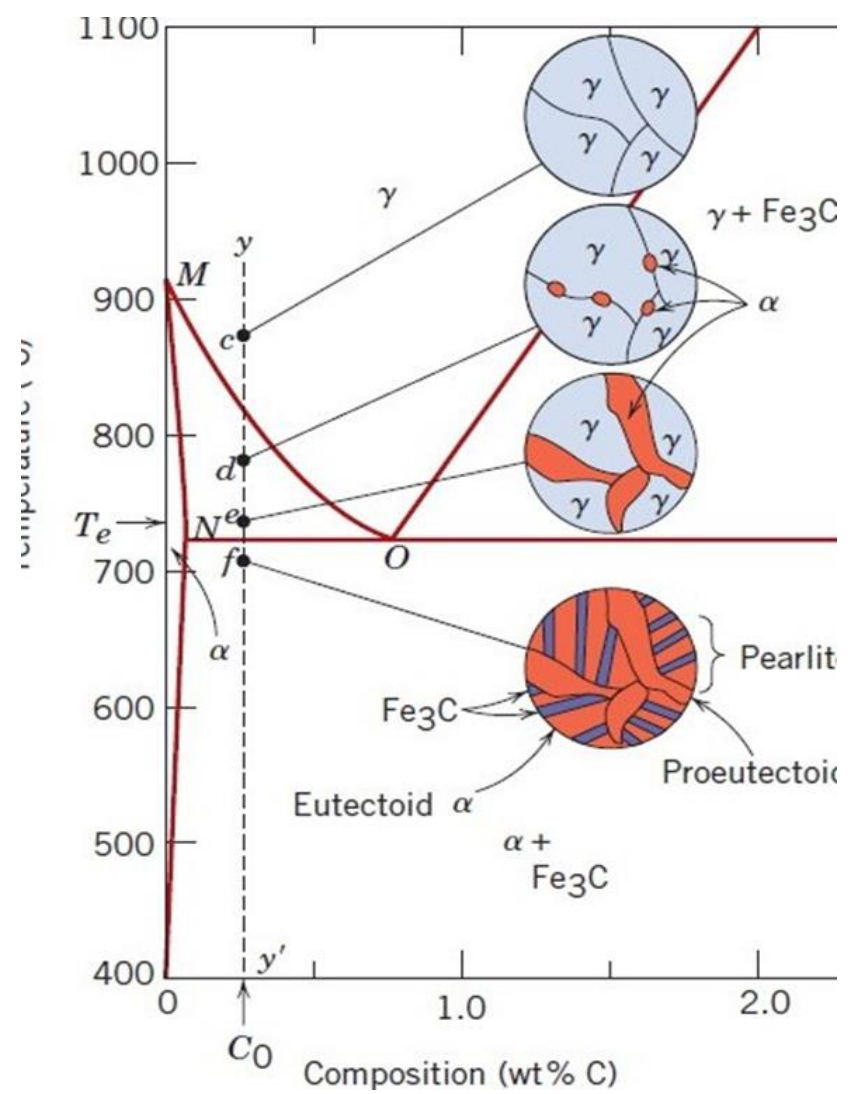
- The microstructure for eutectoid steel that is slowly cooled through the eutectoid temperature consists of alternating layers or lamellae of the two phases (α and Fe_3C) that form simultaneously during the transformation.
- This microstructure, point b, is called pearlite because it has the appearance of mother of pearl when viewed under the microscope at low magnifications.
- The pearlite exists as grains, often termed “colonies”; within each colony the layers are oriented in essentially the same direction, which varies from one colony to another.
- The thick light layers are the ferrite phase, and the cementite phase appears as thin lamellae most of which appear dark. Mechanically, pearlite has properties intermediate between the so ductile

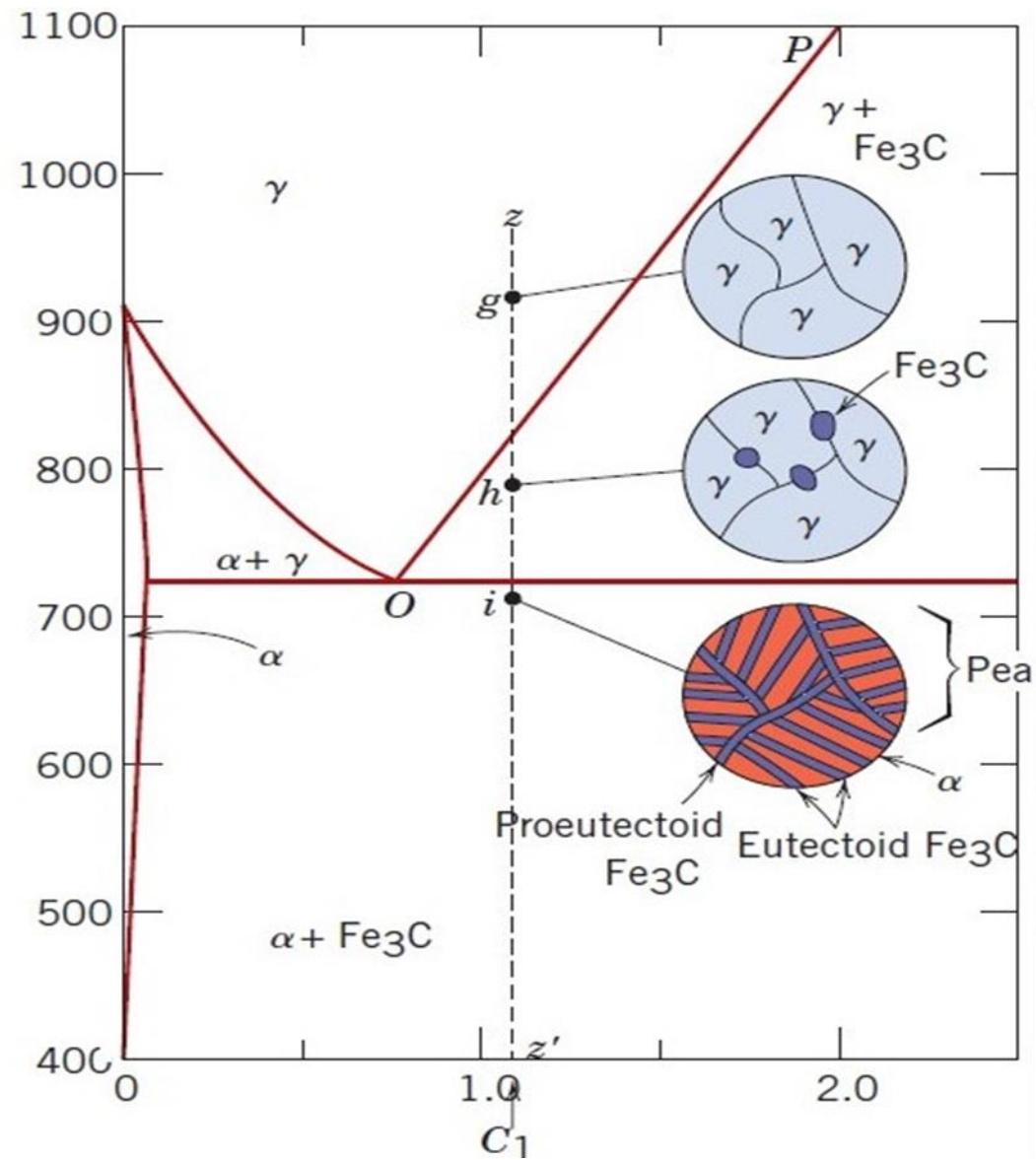


- The alternating α and Fe_3C layers in pearlite form as such for the same reason that the eutectic structure forms, because the composition of the parent phase [in this case austenite (0.76 wt% C)] is different from either of the product phases [ferrite (0.022 wt% C) and cementite (6.7 wt% C)], and the phase transformation requires that there be a redistribution of the carbon by diffusion.
- The directions of carbon diffusion are indicated by arrows. Carbon atoms diffuse away from the 0.022 wt% ferrite regions and to the 6.7 wt% cementite layers, as the pearlite extends from the grain boundary into the unreacted austenite grain. The layered pearlite forms because carbon atoms need diffuse only minimal distances with the formation of this structure.

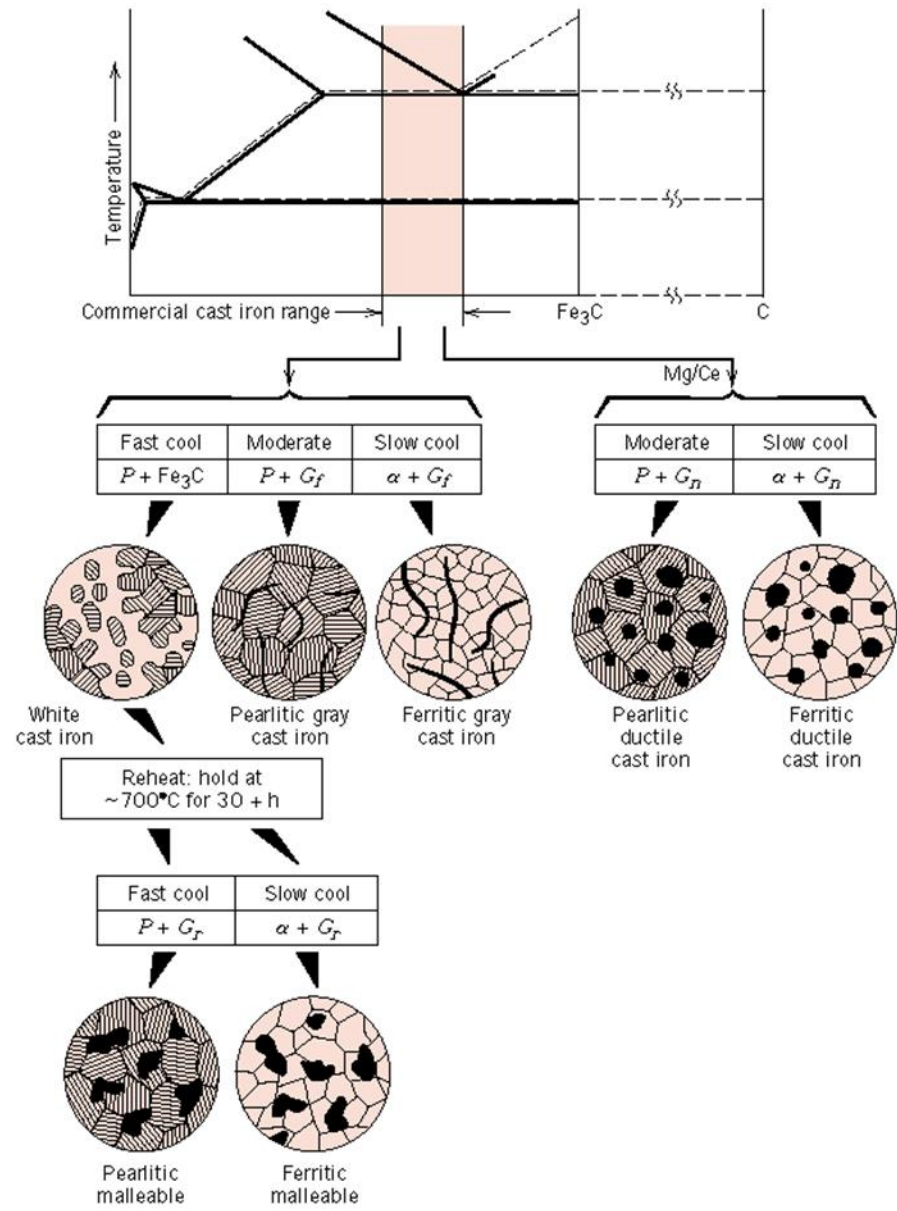


- Consider a composition C_0 to the left of the eutectoid, between 0.022 and 0.76 wt% C; this is termed a ***hypoeutectoid*** (less than eutectoid) alloy.
- Cooling an alloy of this composition is represented by moving down the vertical line yy'
- At about 875°C, point c, the microstructure will consist entirely of grains of the γ phase, as
- In cooling to point d, about 775°C, which is within the $\alpha+\gamma$ phase region, both these phases will coexist as in the schematic microstructure.
- Most of the small α particles will form along the original γ grain boundaries.





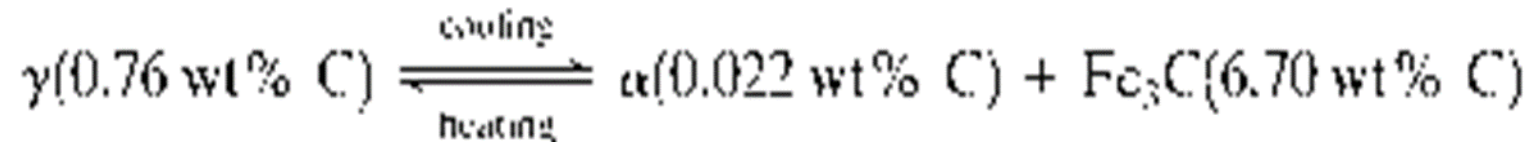
- On the other hand, the change in composition of the austenite is more dramatic, proceeding along the $(\alpha+\gamma) - \gamma$ boundary, line MO, as the temperature is reduced.
- • Cooling from point d to e, just above the eutectoid but still in the $\alpha+\gamma$ region, will produce an increased fraction of the α phase and a microstructure similar to that also shown: the α particles will have grown larger. The α phase will contain 0.022 wt% C, while the γ phase will be of the eutectoid composition, 0.76 wt% C.
- • As the temperature is lowered just below the eutectoid, to point f, all the γ phase that was present at temperature T_e (and having the eutectoid composition) will transform to pearlite.
- • There will be virtually no change in the α phase that existed at point e in crossing the eutectoid temperature—it will normally be present as a continuous matrix phase surrounding the isolated pearlite colonies. Thus the ferrite (α) phase will be present both in the pearlite and also as the phase that formed while cooling through the $\alpha+\gamma$ phase region.
- • The ferrite that is present in the pearlite is called ***eutectoid ferrite***, whereas the other, that formed above T_e , is termed ***proeutectoid*** (meaning pre- or before eutectoid) ferrite.



ISOTHERMAL TRANSFORMATION DIAGRAMS (TTT DIAGRAMS)

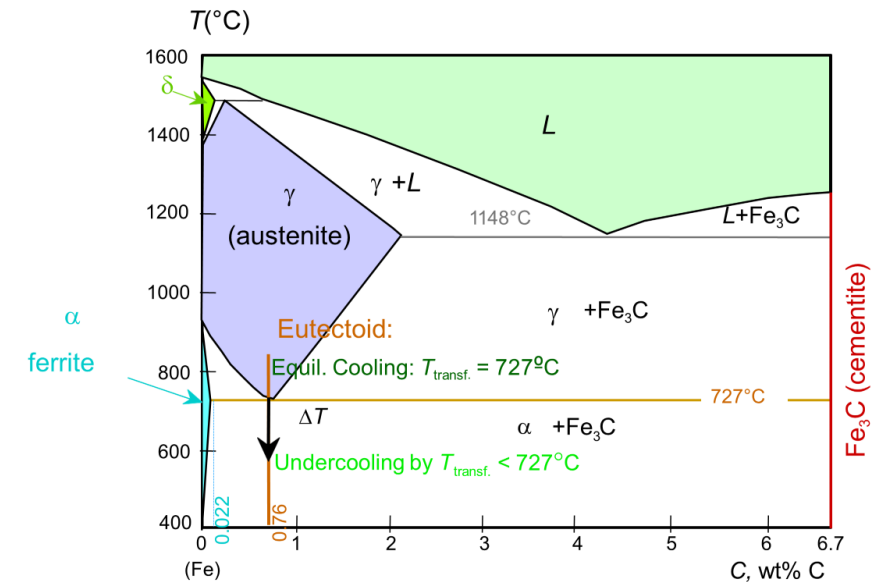
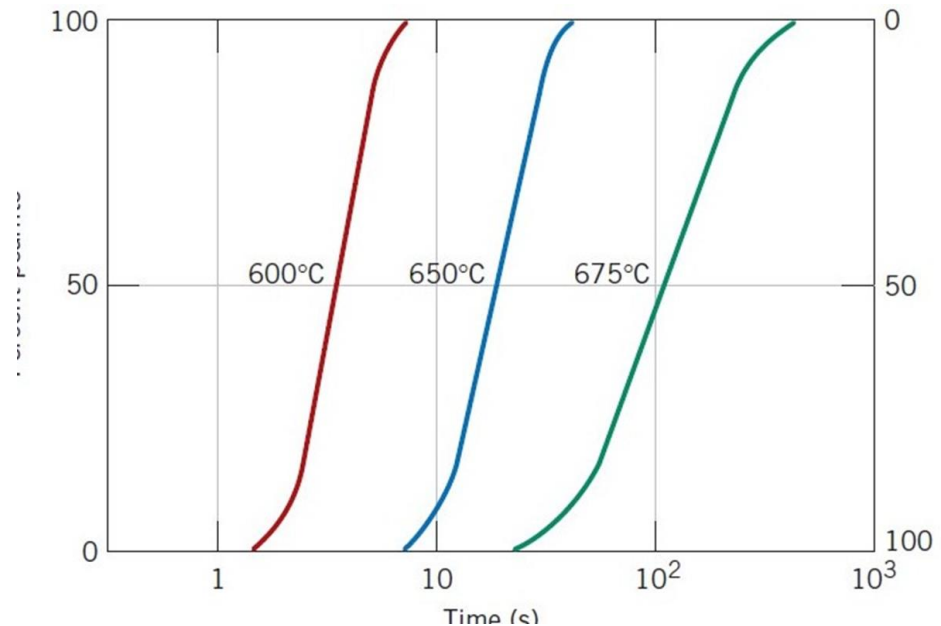
- There are several restrictions for the phase diagram, even though they are useful in the interpretations of metallic and ceramic materials.
- It presents only compositional data, limited information on the structural form.
- Phase diagram gives only equilibrium states.
- Alloys are rarely cooled at very slow rates or at equilibrium cooling.
- To overcome these difficulties isothermal transformation diagrams are used.

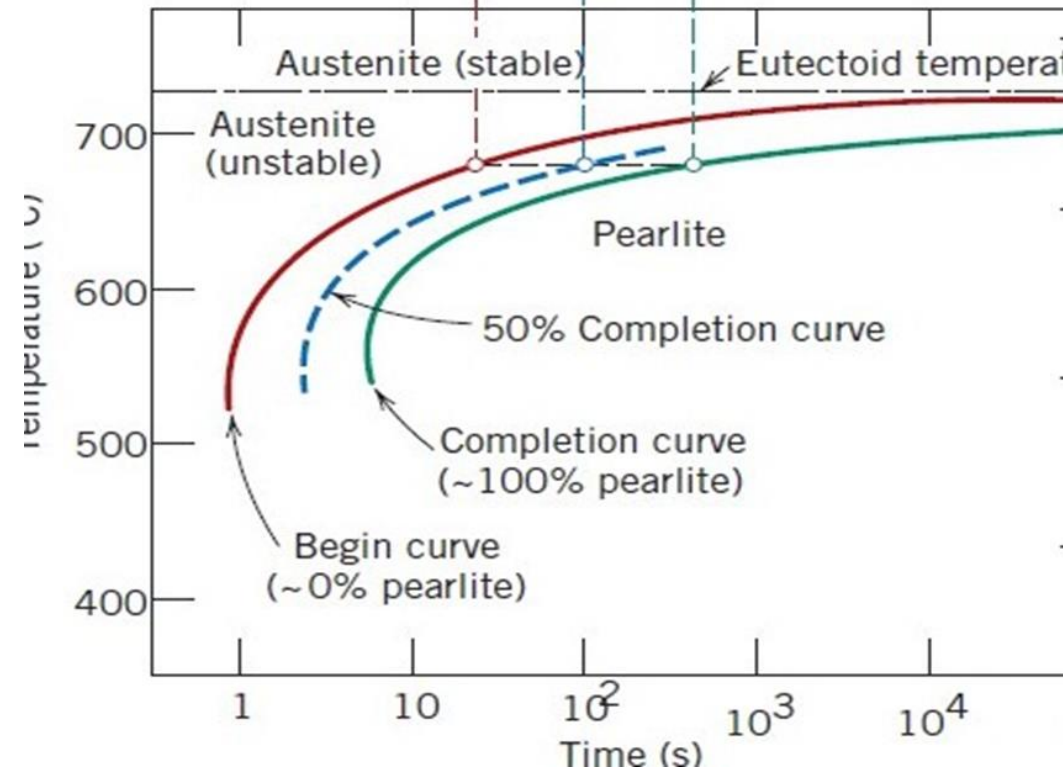
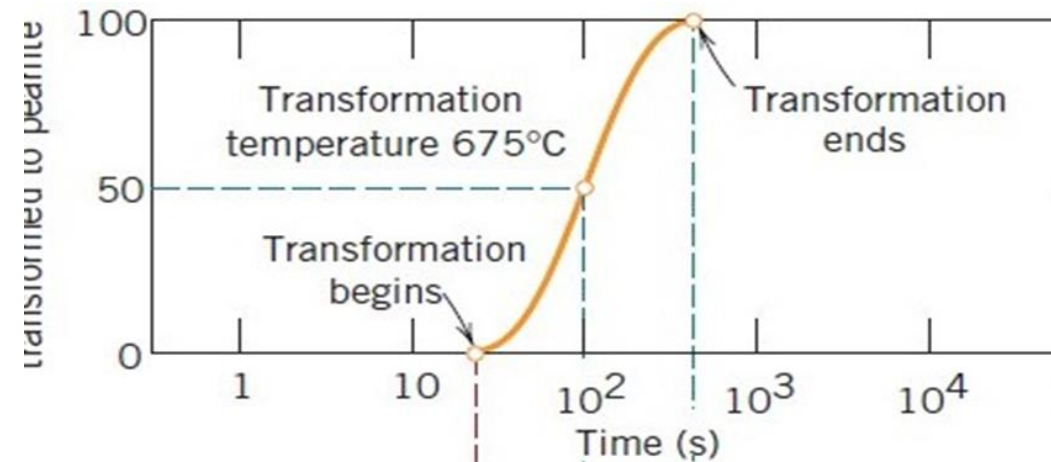
- Temperature and cooling rate play a very important role in austenite to pearlite transformation in steel



- Upon cooling, austenite, having an intermediate carbon concentration, transforms to *a ferrite phase, having a much lower carbon content*, and also *cementite, with a much higher carbon concentration*. Pearlite is one microstructural product of transformation

- Temperature plays an important role in the rate of the austenite-to- pearlite transformation.
- The temperature dependence for an iron-carbon alloy of eutectoid composition is indicated in **Figure**, which plots S-shaped curves of the percentage transformation versus the logarithm of time at three different temperatures.
- For each curve, data were collected after rapidly cooling a specimen composed of 100% austenite (heated $>727^{\circ}\text{C}$, depending on the composition of Carbon) to the indicated temperature; maintained at this constant temperature during the course of the reaction.
- The time required for austenite to pearlite transformation is noted. As the temperature to which the samples are cooled increases, the time required for transformation (t) increases (i.e., $t_{600} < t_{650} < t_{675}$)

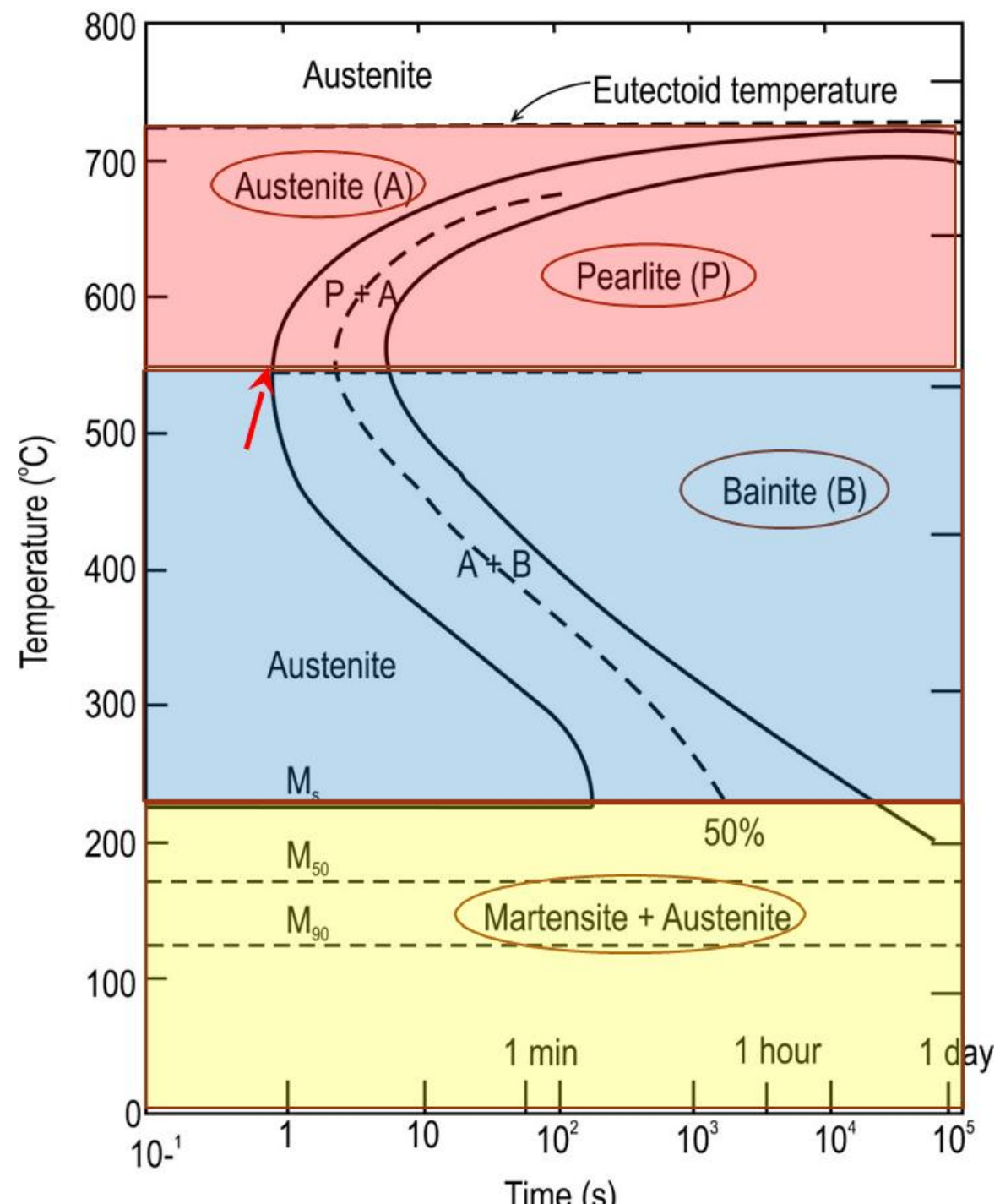




TTT Diagram

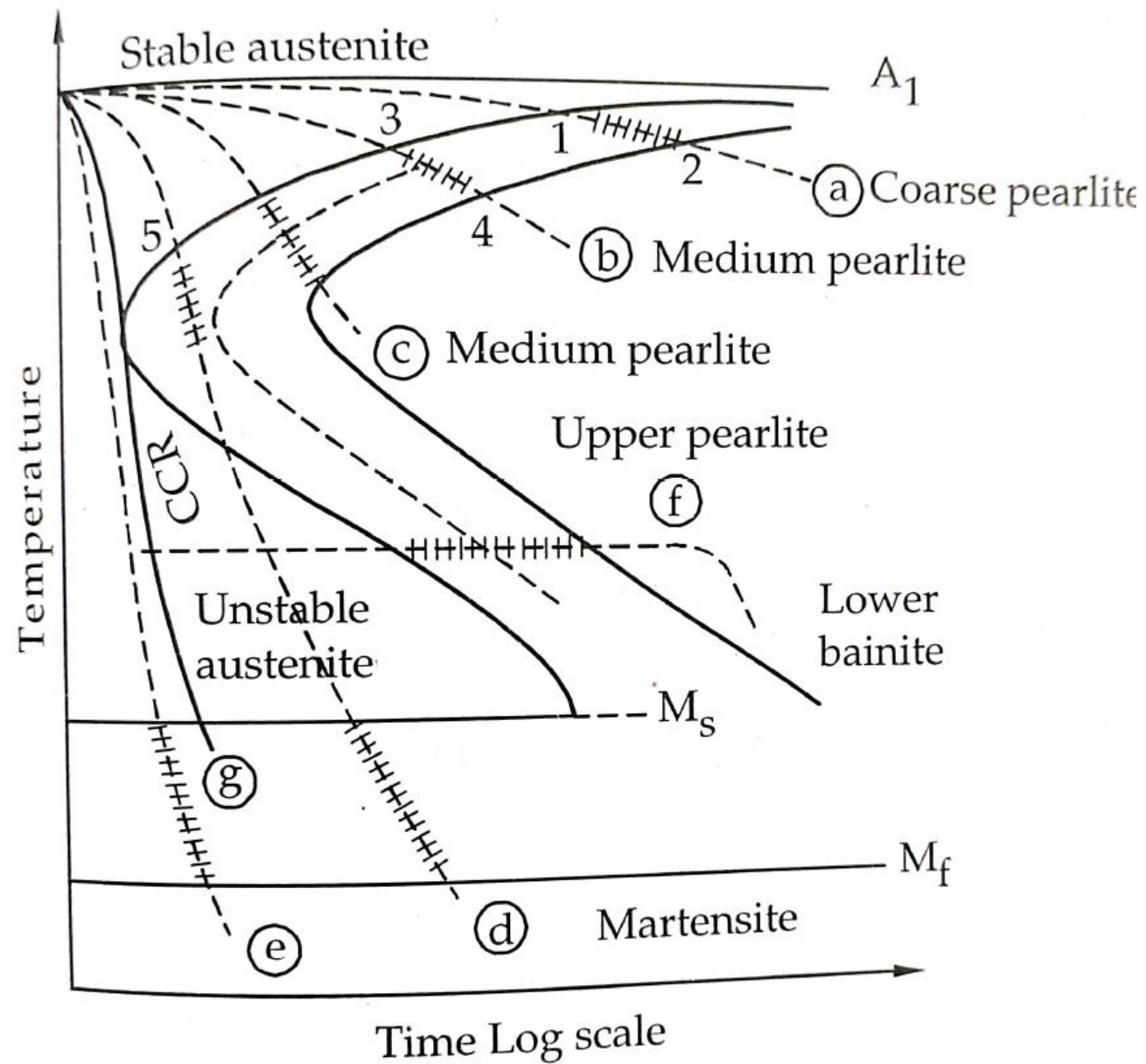
- TTT diagram stands for “time-temperature- transformation” diagram.
- It is also called isothermal transformation diagram.
- TTT diagrams give the kinetics of isothermal transformations.
- The iron-iron carbide equilibrium diagram is having little value in the study of the steels cooled under non equilibrium conditions.
- It has been found in the study of steel that, Austenite is unstable below the lower critical temperature A1.
- Below A1 temperature austenite may transform to pearlite, bainite or martensite.
- It is necessary to know that, How these transformation occurred.

- This diagram indicates the phases existing in steel of a specific composition at various temperatures and times.
- It is helpful to obtain desired microstructure and hence to design properties of the alloy.



Continuous Cooling Transformation (CCT) Diagram

- A cooling curve is drawn with the experimentally determined data by placing a thermocouple at a definite location in steel sample and then measuring the variation of temperature with time.
- Since the coordinates of the TTT and CCT diagram are same those for a cooling curve.
- It is possible to superimpose variation cooling curve on them.



- In interpreting this diagram, note first that the eutectoid temperature [727°C] is indicated by a horizontal line; at temperatures above the eutectoid and for all times, only austenite will exist.
- The austenite-to-pearlite transformation will occur only if an alloy is super cooled to below the eutectoid, the time necessary for the transformation to begin and then end depends on temperature.
- The start and finish curves are nearly parallel, and they approach the eutectoid line asymptotically. To the left of the transformation start curve, only austenite (which is unstable) will be present, whereas to the right of the finish curve, only pearlite will exist.
- In between, the austenite is in the process of transforming to pearlite, and thus both microconstituents will be present.
- The transformation rate at some particular temperature is inversely proportional to the time required for the reaction to proceed to 50% completion (to the dashed line in **Figure**).
- That is, the shorter the time, the higher is the rate. Thus, at temperatures just below the eutectoid (corresponding to just a slight degree of undercooling) very long times (on the order of 10^5 s) are required for the 50% transformation, and therefore the reaction rate is very slow.
- The transformation rate increases with decreasing temperature such that at 540°C only about 3 s is required for the reaction to go to 50% completion.

- At temperatures just below the eutectoid, relatively thick layers of both the α -ferrite and Fe_3C phases are produced; this
 - microstructure is called **coarse pearlite**,
- At these temperatures, diffusion rates are relatively high, such that during the transformation carbon atoms can diffuse relatively long distances, which results in the formation of thick lamellae.
- With decreasing temperature, the carbon diffusion rate decreases, and the layers become progressively thinner.
- The thin-layered structure produced is termed **fine pearlite**.
- Fine pearlite leads to better hardness and strength

- **Bainite**

- In addition to pearlite, other microconstituents that are products of the austenitic transformation exist; one of these is called **bainite**.
- The microstructure of **bainite** consists of ferrite and cementite phases, and thus diffusional processes are involved in its formation.
- Bainite forms as feather or needles or plate like cementite

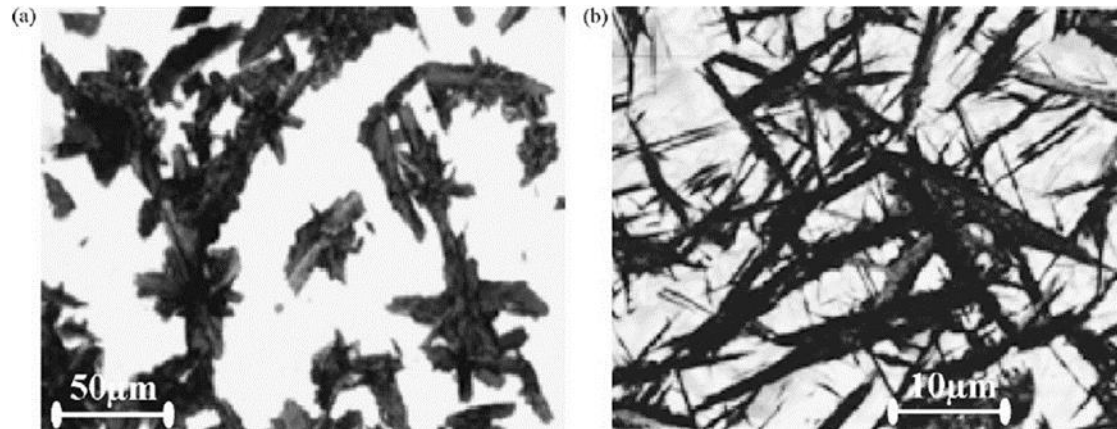
Differentiate between bainite and pearlite

Pearlite has a lamellar microstructure with alternating layers of ferrite and cementite, yielding intermediate strength and ductility.

Bainite displays a non-lamellar acicular structure with a fine mixture of ferrite and cementite, resulting in higher strength but less ductility compared to pearlite.

- Bainite leads to more strength and hardness compared to pearlite.
- This is due to smaller ferrite particles.
- There will be an increase in strength associated decrease in ductility

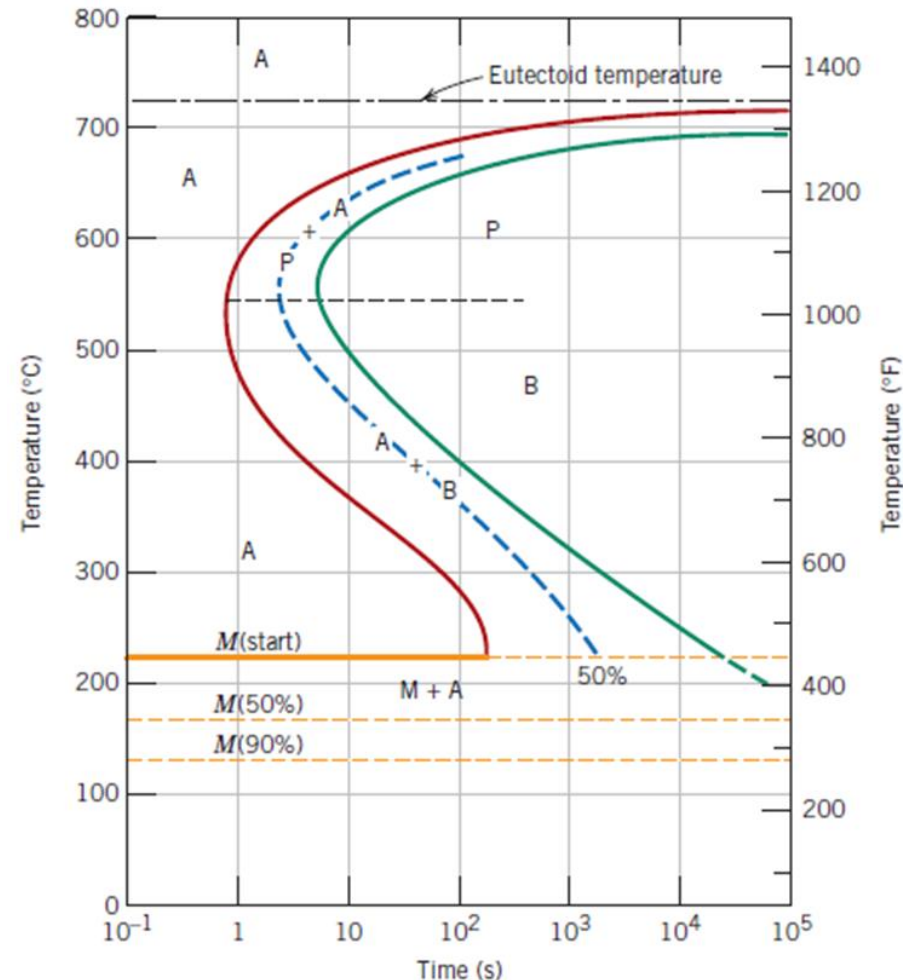
- The time-temperature dependence of the bainite transformation can be represented on the isothermal transformation diagram.
- It occurs at temperatures below those at which pearlite forms; begin-, end-, and half-reaction curves are just extensions of those for the pearlitic transformation.
- The isothermal transformation diagram for an iron-carbon alloy of eutectoid composition that has been extended to lower temperatures.
- All three curves are C-shaped and have a “nose” at point *N*, where the rate of transformation is a maximum.
- Pearlite forms above the nose [i.e., over the temperature range of about 540 to 727 °C, at temperatures between about 215 and 540 °C, bainite is the transformation product.
- Temperatures just below the nose TTT curve is called *upper bainite* with feathery cementite appearance.
- At lower temperatures, the cementite grains become too fine and an acicular needle like pattern is observed. This is lower bainite which is also called *acicular bainite*.



Upper bainite

Acicular bainite

- On rapid cooling (quenching) to low temperatures ($<215\text{ }^{\circ}\text{C}$), the austenite transforms to another microstructure called Martensite.
- Martensite forms when the rate of cooling is rapid enough to prevent diffusion.
- The formation of martensite grains nucleate and grow at a very rapid rate.



- Being a nonequilibrium phase, martensite does not appear on the iron-iron carbide phase diagram.
- The austenite-to-martensite transformation is, however, represented on the isothermal transformation diagram.
- The martensitic transformation is diffusionless and instantaneous, it is not depicted in this diagram as the pearlitic and bainitic reactions.
- The beginning of this transformation is represented by a horizontal line designated $M(\text{start})$. Two other horizontal and dashed lines, labeled $M(50\%)$ and $M(90\%)$, indicate percentages of the austenite-to-martensite transformation.
- The temperatures at which these lines are located vary with alloy composition but, nevertheless, must be relatively low because carbon diffusion must be low.
- The horizontal and linear character of these lines indicates that the martensitic transformation is independent of time; it is a function only of the temperature to which the alloy is quenched or rapidly cooled.
- Transformation to martensite an **athermal transformation**.

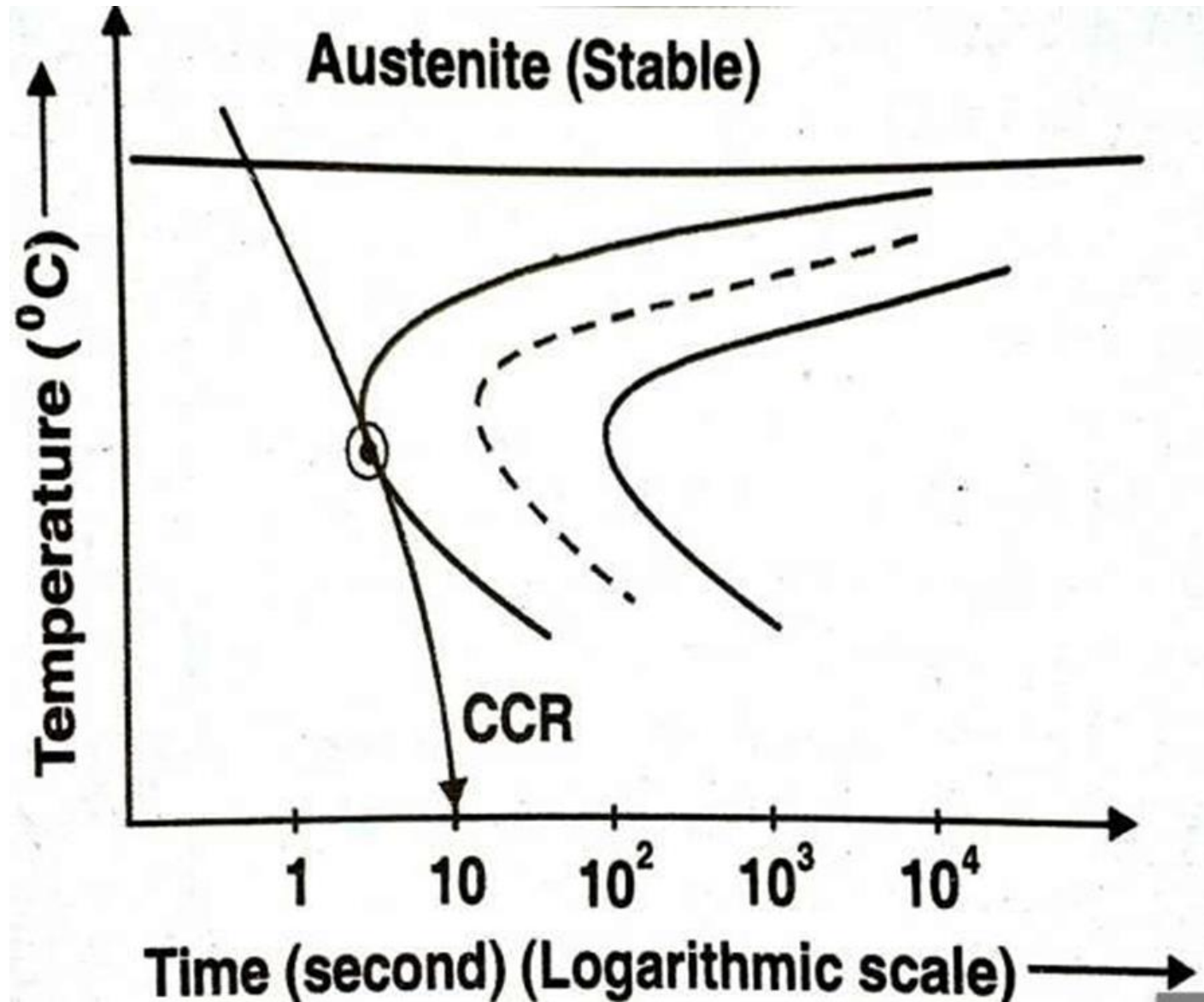
- In martensite transformation, a sudden orientation of carbon and iron atoms from FCC solid solution (Austenite) into a body centered tetragonal (BCT) solid solution happens here.
- Compared to other microstructures observed in steel, martensite is the strongest and hardest.
- These properties are due to the presence of Interstitial carbon atoms.
- The interstitial impurities acts as barriers to dislocation motion.



Critical Cooling Rate

- The minimum rate of continuous cooling just sufficient to prevent undesired transformations.
- For steel, the slowest rate at which it can be cooled from above the upper critical temperature to prevent the decomposition of austenite at any temperature above when Martensite forms (M_s), around 315°C.
- When cooling curve tangent to the nose of TTT curve, it is called as critical cooling rate(CCR).
- CCR can be defined by two ways 'The slowest cooling rate at which unstable austenite can be transformed into martensite.'
- 'The faster cooling rate or within minimum time at which unstable austenite can be transformed into perlite.'

Critical Cooling Rate



Factors influencing Cooling Rate

- ✓ Carbon content.
- ✓ Austenite temperature.

1) Carbon content:-

- With higher carbon content or alloying element, decrease critical cooling rate and shift TTT curve towards right.
- It rated the transformation of austenite to perlite or bainite.

2) Austenite temperature:-

- Higher the austenitic temperature, it will coarsen the structure and make it softer.
- Higher austenitic temperature gives lower critical cooling time and increase CCR.

